Gaussians are not local maximizers of the endpoint Strichartz inequality in dimension five

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Abstract

Remark 1.3(c) of Gonçalves, Negro and Oliveira e Silva [1] leaves open whether Gaussians are local maximizers at the threshold exponent $r = \rho_d$, and notes that for $d = 5$ this is the Keel–Tao endpoint $L_t^2 L_x^{10/3}$. We settle the case $d = 5$: Gaussians are not local maximizers there. We give explicit real, positive, radial perturbations f_ε of the Gaussian whose Strichartz quotient exceeds the Gaussian value by a relative amount greater than $11\varepsilon^4$ for every $0 < \varepsilon \le 10^{-3}$. The proof uses two-sided Taylor bounds and polynomials with exact rational coefficients; it does not rely on a formal expansion of the fractional norm. An exact-arithmetic checker for the rational part of the certificate is sent with the note as a separate file. We do not determine the sharp constant or the global maximizers, and the threshold cases $d \le 4$ remain open.

1 Result

Let $\Delta = \sum_{j=1}^5 \partial_j^2$, $S(t) = e^{-it\Delta/2}$ on $\mathbb{R}^5$, and $g(x) = e^{-|x|^2/2}$. All norms use Lebesgue measure, and $\|\cdot\|_{2,10/3}$ denotes the norm of $L_t^2(\mathbb{R}; L_x^{10/3}(\mathbb{R}^5))$. Put $s = |x|^2$ and

$$P(s) = s^2 - 7s + \frac{35}{4},$$
$$Q(s) = s^4 - 22s^3 + \frac{297}{2}s^2 - \frac{693}{2}s + \frac{3465}{16}, \qquad \beta = \frac{194}{683}.$$

Theorem 1. *For every real $0 < \varepsilon \le 10^{-3}$, the real-valued radial function*

$$f_\varepsilon(x) = g(x)\big[1 + \varepsilon P(|x|^2) + \beta\varepsilon^2 Q(|x|^2)\big]$$

is positive and Schwartz, and

$$\frac{\|S(t)f_\varepsilon\|_{2,10/3}^2}{(3/5)^{3/2}\|f_\varepsilon\|_2^2} - 1 > 11\varepsilon^4.$$

For comparison, with $\beta = 0$ the same relative gain is below $-46\varepsilon^4$ for every $0 < \varepsilon \le 10^{-3}$, and g itself gives 0.

The constant $(3/5)^{3/2}$ is the Gaussian value of the quotient. It follows by direct integration of the propagated Gaussian (A.2) of [1]: $\|S(t)g\|_{10/3}^2 = (3\pi/5)^{3/2}(1+t^2)^{-1}$, so $\|S(t)g\|_{2,10/3}^2 = \pi(3\pi/5)^{3/2}$, while $\|g\|_2^2 = \pi^{5/2}$.

Write $\delta[f] = (3/5)^{3/2}\|f\|_2^2 - \|S(t)f\|_{2,10/3}^2$ for the deficit $\delta_{2,10/3}$ of [1]; it is negative exactly when the relative gain in Theorem 1 is positive. By Definition 1.1 of [1] (whose ε we call a), Gaussians

are local maximizers if there is $a > 0$ such that $\delta[f] \geq 0$ for every f with $\mathrm{dist}(f, \mathrm{G}) < a\|f\|_2$, where the distance is in L^2 to the Gaussian orbit G. To show they are not, it is enough to give, for every $a > 0$, some f in that neighbourhood with negative deficit. Since $\langle g, f_\varepsilon - g\rangle = 0$ (the modes gP and gQ are orthogonal to g), we have $\|f_\varepsilon - g\|_2^2 = \|g\|_2^2\,(N(\varepsilon) - 1)$, with N as in (2). Hence

$$\frac{\mathrm{dist}(f_\varepsilon, \mathrm{G})}{\|f_\varepsilon\|_2} \leq \frac{\|f_\varepsilon - g\|_2}{\|f_\varepsilon\|_2} = \sqrt{\frac{N(\varepsilon) - 1}{N(\varepsilon)}} \longrightarrow 0,$$

while Theorem 1 gives $\delta[f_\varepsilon] = (3/5)^{3/2}\|g\|_2^2\,(N - B) < 0$. So Gaussians are not local maximizers at $d = 5$, $r = 10/3$.

The effect is small. At $\varepsilon = 10^{-3}$ the relative gain is about 1.15×10^{-11}. The interval is not optimized. In uncertified numerical quadrature the formal ε^6 coefficient of $B - N$ is about -1935 (about -2137 for the relative gain), and the gain changes sign between $\varepsilon = 0.07$ and 0.08. None of this is used in the proof.

2 Why this direction, and the quartic coefficient

In the notation of Section 2 of [1], $P = 2L_2^{(3/2)}(s)$ and $Q = 24L_4^{(3/2)}(s)$, so gP and gQ are multiples of the radial modes $f_{2,0}$ and $f_{4,0}$. The number operator $\mathcal{K} = (-\Delta + |x|^2 - 5)/2$ (the operator $-(\mathcal{H} + d)/2$ of Lemma 2.4 and Proposition 2.3 of [1], with $\mathcal{H} = \Delta - |x|^2$) acts on $gR(s)$ by $R \mapsto -2sR'' + (2s - 5)R'$, and $\mathcal{K}P = 4P$, $\mathcal{K}Q = 8Q$ in this sense. By Remark 3.2 of [1], at $d = 5$, $r = \rho_5 = 10/3$ the kernel of the second variation on $(T_g\mathrm{G})^\perp$ is the complex span of $f_{2,0}$, from the factorization $(1 - p_{2,0}(\lambda))/(1 - \lambda) = \frac{25}{4}(\lambda - \frac{3}{5})(\lambda + \frac{1}{5})$ at $\lambda = 2/r$. The linear directions gP and igP are related by the oscillator evolution $e^{iT_0\mathcal{K}}$ with $T_0 = \pi/8$, which preserves the quotient, so the real direction covers the kernel at first order. (The same shift sends Q to $-Q$, so this is a statement about directions, not about the corrected family.) The vanishing of the ε^2 term below is consistent with this; the certificate checks it directly, so the proof does not rely on Remark 3.2.

The lens formula. By Propositions 2.2 and 2.3 of [1] with $q = 2$, for every f,

$$\|S(t)f\|_{2,10/3}^2 = \frac{1}{2}\int_{-\pi}^{\pi} \|e^{iT\mathcal{K}}f\|_{L^{10/3}}^2\,dT.$$

Relative to g, the two modes acquire phases e^{4iT} and e^{8iT}. The integrand depends on T only through $\theta = 4T$ modulo 2π, so the average over T equals the average over $\theta \in [0, 2\pi]$. The normalized squared output is therefore

$$B(\varepsilon) = \frac{\|S(t)f_\varepsilon\|_{2,10/3}^2}{\|S(t)g\|_{2,10/3}^2} = \mathrm{avg}_{0\leq\theta\leq 2\pi}\left[\mathbb{E}\left|1 + Ae^{i\theta} + De^{2i\theta}\right|^{10/3}\right]^{3/5}. \tag{1}$$

Here $A = \varepsilon P(s)$, $D = \beta\varepsilon^2 Q(s)$, $\mathrm{avg} = (2\pi)^{-1}\int_0^{2\pi}$, and $\mathbb{E}$ is expectation over $s = |y|^2$ for $y \in \mathbb{R}^5$ with density proportional to $e^{-5|y|^2/3}$. Thus $s \sim \mathrm{Gamma}(5/2, \mathrm{scale}\ 3/5)$ and $\mathbb{E}s^j = (5/2)_j(3/5)^j$. The input norm uses the weight g^2, under which $s \sim \mathrm{Gamma}(5/2, 1)$ and P, Q, PQ have mean zero. So

$$N(\varepsilon) = \frac{\|f_\varepsilon\|_2^2}{\|g\|_2^2} = 1 + \frac{35}{2}\varepsilon^2 + \frac{10395}{2}\beta^2\varepsilon^4, \tag{2}$$

and the gain in Theorem 1 is $(B - N)/N$. We write N_0 and B_0 for N and B with $\beta = 0$.

The quartic. Replacing θ by $\theta+\pi$ sends A to $-A$ and fixes D, so B is even in ε and the odd coefficients of $B-N$ vanish. The ε^2 coefficient vanishes because P is in the kernel. For fixed real β, the ε^4 coefficient of $B-N$ (so the normalization N is already subtracted) is

$$C(\beta)=\frac{-736344+6453216\beta-11359656\beta^2}{15625}=\frac{984312}{85375}-\frac{11359656}{15625}\Big(\beta-\frac{194}{683}\Big)^2. \tag{3}$$

It is maximal at $\beta=194/683$, where $C=984312/85375\approx 11.53$, and it is positive for β between its roots, about 0.158 and 0.410. In contrast $C(0)=-736344/15625\approx -47.13$: along the kernel direction alone the Gaussian wins at fourth order, and the correction in the mode $f_{4,0}$ reverses the sign. The $\beta=0$ statement in Theorem 1 illustrates this; it is not needed for the conclusion.

Remark. Optimizing the formal fourth-order coefficient over real multiples of the radial eigenvalue-8 mode gives 984312/85375. Whether this is the quartic of a Lyapunov–Schmidt reduction of the full functional would need the spectral gap of Remark 3.2 together with a treatment of complex corrections and of the tangent directions; we do not pursue it. The theorem does not use this remark.

The coefficient calculation alone does not justify the inequality. The global bounds of the next two sections do, and they recover $C(194/683)$ exactly as the coefficient v_4.

3 Two-sided Taylor bounds

Let $T_n^a(x)=\sum_{j=0}^n \binom{a}{j}x^j$. For every $x\ge -1$,

$$T_5^{5/3}(x)\le (1+x)^{5/3}\le T_5^{5/3}(x)+\frac{14}{729}x^6. \tag{4}$$

Indeed, with $m=5/3$,

$$\frac{(1+x)^m-T_5^m(x)}{x^6}=6\binom{m}{6}\int_0^1(1-\tau)^5(1+\tau x)^{m-6}\,d\tau,\qquad 6\binom{m}{6}=\frac{70}{2187}>0.$$

Since $m-6=-13/3<0$, the right side is positive and decreasing in x. Its supremum over $x>-1$ is the limit $x\to -1^+$, which equals $-T_5^{5/3}(-1)=14/729$ (by monotone convergence; the limiting integrand is a multiple of $(1-\tau)^{2/3}$). At $x=0$ both sides of (4) equal 1, and $x=-1$ follows by continuity.

Write $c=\cos\theta$ and

$$Z=A^2+D^2-2D+2A(1+D)c+4Dc^2.$$

Then exactly $1+Z=|1+Ae^{i\theta}+De^{2i\theta}|^2$, so $Z\ge -1$ everywhere, even where εP or $\varepsilon^2 Q$ is large. Let

$$L=\mathbb{E}\,T_5^{5/3}(Z),\qquad U=\mathbb{E}\left[T_5^{5/3}(Z)+\frac{14}{729}Z^6\right].$$

By (4), $L\le M\le U$ for the actual inner moment $M=\mathbb{E}|1+Ae^{i\theta}+De^{2i\theta}|^{10/3}$. Both are polynomials in c,ε with rational coefficients and constant term one.

For $H=L$ (with $\beta=194/683$) or $H=U$ (with $\beta=0$), write $H-1=\sum c_{jk}c^j\varepsilon^k$ and put

$$R_H(\varepsilon)=\sum|c_{jk}|\varepsilon^k,\qquad \eta=10^{-3},\qquad \rho_H=R_H(\eta).$$

Since $|c|\le 1$ and $\varepsilon\ge 0$, $|H-1|\le R_H(\varepsilon)\le\rho_H$ for $0\le\varepsilon\le\eta$. Exact arithmetic gives $0.0047161411<\rho_L<0.0047161412$ (for $\beta=194/683$) and $0.0047106772<\rho_U<0.0047106773$ (for $\beta=0$), both

below 1/100. Hence $H \geq 1-\rho_H > 0$, so $z \mapsto z^{3/5}$ may be applied to $L \leq M \leq U$. Taylor's theorem with Lagrange remainder, $(1+z)^{3/5} - T_4^{3/5}(z) = \binom{3/5}{5}(1+\xi)^{-22/5} z^5$ with $|\xi| \leq |z| \leq \rho_H$, then gives

$$\left| H^{3/5} - T_4^{3/5}(H-1) \right| \leq 2 \left| \binom{3/5}{5} \right| R_H(\varepsilon)^5, \tag{5}$$

since $(1-\rho_H)^{22} > 1/32$ makes $(1+\xi)^{-22/5} \leq (1-\rho_H)^{-22/5} < 2$. The error is kept as a polynomial in ε starting at degree five. This is what proves the result for every small ε, not only at η. Time averages are exact: $\operatorname{avg} c^{2j} = \binom{2j}{j}/4^j$ and $\operatorname{avg} c^{2j+1} = 0$. So, for each of the two families, the normalized squared norm is bounded by an explicit polynomial with rational coefficients, with no quadrature and no omitted spatial tail.

4 The rational certificate

For $\beta = 194/683$ let

$$V = \operatorname{avg} T_4^{3/5}(L-1) - 2 \left| \binom{3/5}{5} \right| R_L(\varepsilon)^5 - N.$$

Then

$$B = \operatorname{avg} M^{3/5} \;\geq\; \operatorname{avg} L^{3/5} \;\geq\; \operatorname{avg} T_4^{3/5}(L-1) - 2 \left| \binom{3/5}{5} \right| R_L(\varepsilon)^5,$$

so $B - N \geq V$. For this β, Z has degree 4 in ε, so L and R_L have degree 20 and V has degree 100. V has no terms below degree four. The averaged Taylor part $\operatorname{avg} T_4^{3/5}(L-1)$ is even in ε, because every monomial of Z has c-degree congruent to its ε-degree modulo 2 and odd powers of c average to zero. The envelope term $R_L(\varepsilon)^5$ starts at degree five (it does have odd-degree terms, all counted in the tail below). After subtracting N, the constant terms cancel and the ε^2 coefficient is checked to be 0. Its ε^4 coefficient is $v_4 = 984312/85375$. Its tail satisfies

$$0.1187800366 < \sum_{k>4} |v_k| \eta^{k-4} < 0.1187800367 < \frac{3}{25}, \qquad N(\eta) < \frac{1001}{1000}.$$

For $0 < \varepsilon \leq \eta$ we have $\varepsilon^{k-4} \leq \eta^{k-4}$ and $N(\varepsilon) \leq N(\eta)$. So

$$\frac{B-N}{\varepsilon^4} \;\geq\; \frac{984312}{85375} - \frac{3}{25} \;>\; 11N(\eta) \;\geq\; 11N(\varepsilon).$$

The middle comparison is exact: $984312/85375 - 3/25$ exceeds $11N(\eta)$ by about 0.409. Dividing by $N(\varepsilon)$ gives the inequality of Theorem 1.

For $\beta = 0$ let

$$W = \operatorname{avg} T_4^{3/5}(U-1) + 2 \left| \binom{3/5}{5} \right| R_U(\varepsilon)^5 - N_0,$$

so $B_0 = \operatorname{avg} M_0^{3/5} \leq \operatorname{avg} U^{3/5}$ gives $B_0 - N_0 \leq W$. For $\beta = 0$, Z has degree 2 in ε, so U has degree 12 and W has degree 60, again with no terms below degree four, by the same argument. Its lowest-order (ε^4) coefficient is $-736344/15625$, and its absolute tail after division by ε^4 is about $0.13152 < 7/50$. Since $-736344/15625 + 7/50 < -46N_0(\eta)$, we get $B_0 - N_0 < -46\varepsilon^4 N_0(\varepsilon)$, and dividing by $N_0(\varepsilon)$ gives the comparison statement.

family	degree	ε^4 coefficient	tail $\sum_{k>4} \lvert\cdot\rvert \eta^{k-4}$	final comparison
V, $\beta = 194/683$	100	$984312/85375$	$\approx 0.11878 < 3/25$	$v_4 - 3/25 > 11N(\eta)$
W, $\beta = 0$	60	$-736344/15625$	$\approx 0.13152 < 7/50$	$w_4 + 7/50 < -46N_0(\eta)$

5 Positivity

The identities

$$P = (s - 7/2)^2 - 7/2,$$
$$Q = (s - 33/2)^2(s^2 + 11s + 363/4) + (297/2)(s - 7/6)^2 - 98769/4$$

hold for all real s. Since $s = |x|^2 \geq 0$, they give, for $0 \leq \varepsilon \leq \eta$,

$$1 + \varepsilon P + \beta\varepsilon^2 Q \geq 1 - \frac{7}{2}\eta - \frac{98769}{4}\beta\eta^2 = \frac{1351638407}{1366000000} > \frac{49}{50}.$$

The bound on Q is loose but enough. So each f_ε is a positive Schwartz function, which completes the proof. Positivity is part of the statement only; the non-maximality argument does not use it.

Scope

This settles the $d = 5$ case of the equality threshold in Remark 1.3(c) of [1]. With Theorem 1.2(a) there, Gaussians in $d = 5$ are stable local maximizers for $2 < r < 10/3$ and, by Theorem 1, not local maximizers at $r = 10/3$. Together with Theorem 1.2 of [1], it follows that for $d \geq 3$ Gaussians are local maximizers at the Keel–Tao endpoint $(2, 2d/(d-2))$ if and only if $d \geq 6$. Here the cases $d = 3, 4$ of the endpoint come from Theorem 1.2(b) of [1] and its footnote. The thresholds $r = \rho_d$ for $d \leq 4$ remain open. For $d = 3, 4$ the kernel is again the radial mode $(2, 0)$ and the same type of fourth-order calculation can be attempted, but $\rho_3 = 4\sqrt{7} - 6$ and $\rho_4 = 2\sqrt{15} - 4$ are irrational, so the rational certificate does not transfer as it stands, and no outcome is claimed. For $d = 1, 2$ the kernel is the non-radial $(1, 1)$ eigenspace (of complex dimension 2 when $d = 2$). Nor do we determine the sharp constant or the global maximizers.

References

[1] F. Gonçalves, G. Negro and D. Oliveira e Silva, *Gaussians do not always maximize mixed-norm Strichartz inequalities for the Schrödinger equation*, arXiv:2609.11644v1, submitted 10 September 2026 (the only version as of 2 November 2026). Used here: Definition 1.1, Theorem 1.2, Remark 1.3(c), Propositions 2.2–2.3, Lemma 2.4, the Laguerre modes of Section 2, Remark 3.2, and (A.2).

Exact checker

The rational part of the certificate is checked by the script `check_d5_endpoint.py`, sent with this note as a separate file. Its SHA-256 is

```
003a3511a1230bf6763288aeeb2c42d6
91d26a09742a6c7e495978466fa01ca3
```

It needs Python 3.9 or later and only the standard library. Run it as `python3 -I check_d5_endpoint.py`; it takes about 0.2 seconds and prints `ALL EXACT ASSERTIONS PASSED`, or stops at the first failed assertion. It checks the exact algebraic identities and rational bounds used in the certificate: P, Q and their moments, $C(\beta)$, the polynomials L, U, V, W, their tail bounds, the two final comparisons and the positivity bound. It does not check the decimal enclosures quoted in the text, the numerical observations, or the analytic lens and Taylor arguments.